

CS & IT ENGINEERING



Computer Network

Application Layer

Lecture No. - 04



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Recap of Previous Lecture



Topic

HTTP ✓



Topics to be Covered



Topic

HTTP

Topic

Email

SMTP, IMAP(POP)

Topic

FTP

ABOUT ME



Hello, I'm **Abhishek**

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[NAT]

[GATE-2020][1 Mark]



#Q. Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in [non-persistent mode]. The web page contains [text] and [five very small images]. The minimum number of TCP connections required to display the web page completely in your browser is _____.

Ans: 6 TCP connections

#Q. A graphical HTML browser resident at a network client machine Q accesses a static HTML webpage from a HTTP server S. The static HTML page has exactly one static embedded image which is also at S. Assuming no caching, which one of the following is correct about the HTML webpage loading (including the embedded image)?

[GATE-2014, Set-2, 2-Mark]

- (A) Q needs to send at least 2 HTTP requests to S, each necessarily in a separate TCP connection to server S [TRUE for Non-persistent HTTP]
- ✓ (B) Q needs to send at least 2 HTTP requests to S, but a single TCP connection to server S is sufficient [TRUE for persistent HTTP]
- ✗ (C) A single HTTP request from Q to S is sufficient, and a single TCP connection between Q and S is necessary for this FALSE
- ✗ (D) A single HTTP request from Q to S is sufficient, and this is possible without any TCP connection between Q and S FALSE

Ans: B

[MSQ]

[GATE-2023][2 Mark]



#Q. Suppose in a web browser, you click on the `www.gate-2023.in` URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers.

Let RTT denote the round trip time between your local host and (DNS servers) in the DNS hierarchy. The round trip time between the local host and the web server hosting `www.gate-2023.in` is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.

Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

- ☒ A 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
- ☒ B 5 RTTs, in case of persistent HTTP with pipelining.
- ☒ C 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
- ☒ D 6 RTTs, in case of persistent HTTP with pipelining.

Domain Name: www.gate-2023.in

Mode: Iterative Query

DNS Resolver $\longrightarrow$ 3-tier DNS Hierarchy

Time taken for Name Resolution = 3 RTT

Ans: C & D

Non-persistent HTTP with 5 parallel TCP conn



- ① 3RTT $\rightarrow$ Name Resolution
- ② 2RTT $\rightarrow$ for Base HTML File
- ③ 2RTT $\rightarrow$ for 5 objects in parallel
- ④ 2RTT $\rightarrow$ for remain 5 objects in parallel

9RTT

Persistent HTTP with pipeline:-



- ① 3 RTT $\rightarrow$ Name Resolution
- ② 1 RTT $\rightarrow$ for TCP conn.
- ③ 1 RTT $\rightarrow$ for Base HTML file
- ④ 1 RTT $\rightarrow$ All 10 objects in pipeline
[Ignore 10 object transmission time]

6 RTT

#Q. Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

[GATE-2016, Set-2, 1-Mark]

~~(A) HTTP GET request, DNS query, TCP SYN~~

~~(B) DNS query, HTTP GET request, TCP SYN~~

✓ (C) DNS query, TCP SYN, HTTP GET request

~~(D) TCP SYN, DNS query, HTTP GET request~~

Ans: C

Domain Name in URL



① If local DNS cache is Empty

i) DNS query $\longrightarrow$
 $\longleftarrow$ DNS Reply ii)

② Make TCP connection with HTTP server
(web server)

i) TCP Conn. Req. $\xrightarrow{\text{SYN}}$
 $\longleftarrow$ TCP accept ii)

③ Make HTTP Request over the TCP connection

i) HTTP Request MSG $\xrightarrow{\text{GET}}$
 $\longleftarrow$ HTTP Response MSG ii)

#Q. A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

- ✓ (i) HTTP GET request for the index page
- ✓ (ii) DNS request to resolve the web server's name to its IP address
- (iii) HTTP GET request for an image object
- ✓ (iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer?

[GATE-2024, Set-1, 1-Mark]

- ~~(A) (iv), (ii), (iii), (i)~~
- ✓ (C) (ii), (iv), (i), (iii)

Ans: C

- ~~(B) (ii), (iv), (iii), (i)~~
- ~~(D) (iv), (ii), (i), (iii)~~



Topic : HTTP Messages



→ Two types of HTTP Messages :

1. HTTP Request Message

2. HTTP Response Message



Topic : HTTP Request Message ✓



- ASCII (human-readable format)
- HTTP request message contains :

1. Request Line

→ Method : GET / POST / PUT / HEAD

2. Header Lines

3. Body

[MIME]

⇓
(Base-64
Encoding)



Topic : HTTP Request Method



GET Method :

- include user data in URL field of HTTP GET request message

POST Method :

- Web page often includes form input
- User input sent from client to server in entity body of HTTP POST request message



Topic : HTTP Request Method



PUT Method :

- Upload new file (object) to server
- Completely replaces file that exists at specified URL with content in entity body of HTTP PUT request message

HEAD Method :

- Request headers (only)
- if specified URL were requested with an HTTP GET Method



Topic : HTTP Response Message

→ HTTP response message contains :

1. Status Line ✓

→ Contains 'Status code' : [1xx, 2xx, 3xx, 4xx, 5xx]

2. Header Lines

3. DATA



Error
from
server
side



Error
from
client



Topic : HTTP Response Status Code

→ Status code appear in first line in HTTP response message
[sent from server to client]

200 : OK ✓

→ Request succeeded, requested object later in this message

301 : Moved Permanently

→ Requested object moved, new location specified later in this message



Topic : HTTP Response Status Code



400 : Bad Request

→ Request message not understood by server

404 : Not Found

→ Request document not found on this server

505 :

→ HTTP version not supported



Topic : Email

⇒ Client-Server Application



3 major component of Email :

1. User Agent
2. Mail Server
3. SMTP, IMAP(POP)



Topic : User Agent



- Mail Reader / Mail writer
- Composing, editing, reading mail messages
- e.g. Outlook
- Outgoing, incoming messages stored on server



Topic : Mail Server



- Mailbox : Contains incoming message for user
- Message Queue : Outgoing (to be sent) mail messages



User Agent A

Mail Client

SMTP

- ① TCP connection ✓
- ② Mail transfer

Mail Server,
* Multiple MailBox

Message Queue

Mail Server

Mail Client

SMTP

- ① TCP Conn.
- ② Mail transfer

Mail Server

Mail server

Mail Server,
* Multiple Mailbox

Message Queue

IMAP (POP)

- ① TCP connection
- ② Mail download

User Agent B

Mail client



Topic : SMTP



Mail
Server₁

Mail
Server₂

- SMTP : Simple Mail Transfer Protocol
- Protocol between mail servers to send email messages
- Client : Sending mail server
- Server : Receiving mail server



Topic : SMTP

⇒ SMTP uses TCP
(persistent TCP connection)



- Application Layer 'Stateless' protocol
- Uses TCP to reliably transfer email message
[from client mail server to server mail server]
- Port No. 25 (well know port)
- Sending server behave like client to receiving mail server



Topic : SMTP



→ Three phase of transfer :

1. SMTP Handshaking
2. SMTP transfer of messages
3. SMTP closure

} for each mail transfer



Topic : SMTP



- HTTP : Client Pull, SMTP : Client Push
- SMTP uses persistent TCP connections



2 mins Summary



Topic

HTTP ✓

Topic

Email

⇒ SMTP,

[IMAP/POP]
FTP

Topic

~~FTP~~



THANK - YOU

